

Time-Varying Subjective Bad Regime Risk: New Framework for Understanding Asset Prices

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Introductory Setup

Consider the following setup:

- ▶ Coin is tossed every time period
- ▶ If Heads - good times, if Tails - bad times
- ▶ Nature of good/bad times known

Two Types of Models

1. Coin nature or the stochastic rule by which it changes is known
 - ▶ Relative likelihood and expected duration of good/bad times fixed
2. Market participants can never know true coin nature but have time- and horizon-varying beliefs about it
 - ▶ Subjective probability of landing Heads non-stationary and continuously updated

Type 1 vs Type 2

Type 2 more realistic, but Type 1 is the state of literature

Why? Type 2 so far considered empirically and mathematically intractable

- ▶ *”Besides, would Type 2 be materially different?”*

Main Idea of the Paper

Type 2 models can be highly tractable and have unique descriptive and normative uses

Representative Investor

Stochastic Discount Factor is determined by the CRRA representative investor maximizing lifetime utility of consumption

$$\max_{\{C_s^I\}_{s=t}^{\infty}} E_t^I \left[\int_t^{\infty} e^{-\delta(s-t)} \frac{(C_s^I)^{1-\gamma}}{1-\gamma} ds \right]$$

subject to the budget constraint

$$W_t + E_t^I \left[\int_t^{\infty} \frac{M_s}{M_t} Y_s ds \right] \geq E_t^I \left[\int_t^{\infty} \frac{M_s}{M_t} C_s^I ds \right]$$

Equilibrium

By the First Order Conditions and Market Clearing, time- t price of a claim to time- s dividend payout D_s is given by

$$P_t(D_s) = E_t^I \left[e^{-\delta(s-t)} \left(\frac{C_s}{C_t} \right)^{-\gamma} D_s \right].$$

Everything is standard except $E_t^I [.]$ operator.

Regime Beliefs

What is different about $E_t^I [.]$ operator?

- ▶ Evaluated under *arbitrary* regime beliefs $F_{t,s}$ that satisfy dynamic consistency
 - ▶ Main mathematical innovation of the paper
- ▶ $F_{t,s}$ - time-t probability of being in the good regime at future time s
- ▶ The model does not assume or rely on any dynamics of $F_{t,s}$
 - ▶ i.e. True coin nature is unknown and continuously updated in real-time
 - ▶ In type I models, $F_{t,s}$ variation is known and priced in

Asset Price with Single Regime

Asset price expression satisfies

$$P_t = \int_t^{\infty} P_t(D_s) ds$$

In a single-regime setup with only good regime R_1 ($F_{t,s} = 1$ for all s),

$$P_t = \frac{D_t}{r_{R_1} - \mu_{R_1}}$$

Asset Price with Horizon-Invariant Beliefs

With horizon-invariant beliefs ($F_{t,s} = F_t$ for all s),

$$P_t = \frac{D_t}{r_t - \mu_t},$$

$$r_t = F_t r_{R_1} + (1 - F_t) r_{R_2}$$

$$\mu_t = F_t \mu_{R_1} + (1 - F_t) \mu_{R_2}$$

Asset Price with Two-Horizon Beliefs

With two-horizon beliefs ($F_{t,s} = F_t(1)$ for all $s < T_1$ and $F_{t,s} = F_t(2)$ for $s \geq T_1$),

$$P_t = \sum_{p=1}^2 w(p) \frac{D_t}{r_t(p) - \mu_t(p)},$$

$$\sum_{p=1}^2 w(p) = 1.$$

General Expression

More generally,

$$P_t = \sum_{p=1}^N w(p) \frac{D_t}{r_t(p) - \mu_t(p)},$$

$$\sum_{p=1}^N w(p) = 1.$$

Compact General Expression

P_t can be compactly expressed as

$$P_t = \frac{D_t}{\delta_t},$$

where δ_t increases whenever $F_{t,s}$ goes down for any s and vice versa.

General Expression with Macroeconomic and Idiosyncratic Regimes

Formulas so far assumed macroeconomic regimes only. With idiosyncratic regimes as well,

$$P_t = \frac{\tilde{D}_t}{\delta_{\tilde{D},t}} + \frac{e_t}{\delta_{e,t}}.$$

$\tilde{D}_t$ is systematic cash flow level and $\delta_{\tilde{D},t}$ varies due to systematic regime beliefs only

e_t is idiosyncratic cash flow level and $\delta_{e,t}$ varies due to idiosyncratic and systematic regime beliefs

Calibration

- ▶ Good/Bad regimes can be directly calibrated to data
- ▶ Because $F_{t,s}$ is unrestricted by the model, it can be backed out from VIX measures, calibrated based on surveys or modeled using Rational Inattention theory

Explanatory Value

Main source of novelties: $F_{t,s}$ is non-stationary and captures forward looking (abstract) information (even if irreducible to primitive variables).

It qualitatively explains:

- ▶ Market crashes without macroeconomic contractions and fast price recoveries *during* economic downturns
- ▶ Yield curve inversion/reversions without regime shift
- ▶ "Growth Component" of Average Returns
 - ▶ Suppose idiosyncratic bad regime risk is trending downward over the lifetime of a successful firm
 - ▶ Then, its average returns will reflect drift on top of risk premium due to gradually improving risk perceptions

More Topics the Framework Helps With

Other phenomena the framework helps explain:

1. Return predictability linked to the predictability of fundamentals
2. Pro-cyclical gap between VIX and realized volatility
3. Work in progress: Narrative momentum, term structures of risk-free yields and equity risk premia, time-varying factor betas, Efficient Markets Hypothesis...

Normative Uses

1. Valuation using forward-looking expected growth rates and discount rates
 - ▶ Most existing discount rate models are backward-looking and not state-dependent
 - ▶ Because $F_{t,s}$ directly enters rate expressions and can be modeled econometrically, it can make rates forward-looking
 - ▶ Rates updated in real-time based on changes in forward-looking predictive variables
2. Mapping survey expectations to asset pricing models
 - ▶ Most models assume simple beliefs (e.g. Normal) and only calibrate parameters
 - ▶ Or rely on obscure parameters that are hard to calibrate based on surveys
 - ▶ Since $F_{t,s}$ is intuitive, horizon-varying, and unrestricted by model assumptions, it can help map beliefs more precisely

Normative Uses (Continued)

3. Equilibrium modeling of Web3 Cryptocurrency Valuation
 - ▶ Type 1 models have to take a stance on long-term potential of individual currencies
 - ▶ Type 2 models need not since they are non-stationary
4. Alternative to Bayesian portfolio management
 - ▶ No need to make ad-hoc assumptions about priors
 - ▶ Can focus on predictive modeling of regime likelihood instead

Thank You!

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